

Contrast enhancement is done to improve an image for human viewing as well as for image processing tasks. It makes the image features stand out more clearly by making optimal use of colours available on the display or the output device. Contrast manipulation involves changing the range of contrast values in an image.

Segmentation and labeling of objects within a scene is a prerequisite for most object recognition and classification systems. Segmentation is the process of assigning each pixel in the source image to two or more classes. Image segmentation is the process of partitioning the digital image into multiple segments (sets of pixels, also known as super pixels). The goal is to simplify and/or change the contrast representation of an image into something that is more meaningful and easier to analyse. Once the relevant objects have been segmented and labelled, their relevant features can be extracted and used to classify, compare, cluster or recognise desired objects.

Types of images

The MATLAB tool box supports four types of images, namely, grey-level images, binary images, indexed images and RGB images. Brief description of these image types is given below.

Grey-level images. Also referred to as monochrome images, these use 8 bits per pixel, where a pixel value of 0 corresponds to 'black', a pixel value of 255 corresponds to 'white' and intermediate values indicate varying shades of grey. These are also encoded as a 2D array of pixels, with each pixel having 8 bits.

Binary images. These images use 1 bit per pixel, where a 0 usually means 'black' and a 1 means 'white'. These are represented as a 2D array. Small size is the main advantage of binary images.

Indexed images. These images



Fig. 1: Image A1, which has half the image resolution but the same image size



Fig. 2: Image A2, which has one-fourth the image resolution



Fig. 3: Image A3, which has one-eighth the image resolution

are a matrix of integers (X), where each integer refers to a particular row of RGB values in a secondary matrix (map) known as a colour map.

RGB image. In an RGB image, each colour pixel is represented as a triple containing the values of its R, G and B components. In MATLAB, an RGB colour image corresponds to a 3D array of dimensions $M \times N \times 3$. Here 'M' and 'N' are the image's height and width, respectively, and

3 is the number of colour components. For RGB images of class double, the range of values is [0.0, 1.0], and for classes uint8 and uint16, the ranges are [0, 255] and [0, 65535], respectively.

Most monochrome image processing algorithms are carried out using binary or greyscale images.

Image quality

Image quality is defined in terms of spatial resolution and quantisation.

Spatial resolution is the pixel density over the image. The greater the spatial resolution, the more are the pixels used to display the image. Spatial resolution is expressed qualitatively as dots per inch (dpi).

The image resolution can be changed using the `imresize` function. The command `imresize(x,1/2)` halves the image size. This is done by taking a matrix from the original matrix having elements whose row and column indices are even. `imresize(x,2)` means all the pixels are repeated to produce an image of the same size as original, but with half the resolution in each direction.

Other resolution changes can be done by using the desired scaling factors. Command `imresize(imresize(x,1/2),2)` changes the image resolution by half while keeping the image size same. Similarly, command `imresize(imresize(x,1/4),4)` changes the image resolution by one-fourth.

The following set of commands reduces the resolution of image 'Penguins_grey.jpg' by half:

```
A = imread('Penguins_grey.jpg');
A1 = imresize(imresize(A,1/2),2);
imshow(A1)
```

Fig. 1 shows image A1.

```
A = imread('Penguins_grey.jpg');
A2 = imresize(imresize(A,1/4),4);
imshow(A2)
```

Fig. 2 shows image A2.

```
A = imread('Penguins_grey.jpg');
A3 = imresize(imresize(A,1/8),8);
imshow(A3)
```

Fig. 3 shows image A3.